The Suzuki–Weil kernel identity via boundary Clark kernels: a functional-equation diagonal law and a Parseval proof

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Scope. Suzuki [S1, §7.7], [S2, Thm. 1.2] reduces RH to the identity $\int_{\mathbb{R}} \mathfrak{S}_x(z)\overline{\mathfrak{S}_y(z)}\,dz = \pi\, G(x,y)$ for an explicit family $\mathfrak{S}_x \in L^2(\mathbb{R})$. We prove this identity under a single named hypothesis (inner-ness of one explicit meromorphic function), by a route in which every constant is exact and no contour estimate appears: the diagonal normalization turns out to be an *unconditional* consequence of the functional equation alone (Lemma 2), and the off-diagonal vanishing is boundary Clark-kernel orthogonality. Citations: [S1] = arXiv:2606.09096; [S2] = arXiv:2301.00421; [AC] = Ahern–Clark, and standard model-space theory. All statements about what is proven where in [S1,S2] were verified at source.

1 Objects

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, entire, real on $\mathbb{R}$, with $\xi(1-s) = \xi(s)$. Write, for $z \in \mathbb{C}$,

$$A(z) = \xi(\tfrac{1}{2} - iz), \qquad B(z) = \xi'(\tfrac{1}{2} - iz), \qquad E(z) = A(z) + B(z),$$

where $\xi' = d\xi/ds$. For entire F put $F^\sharp(z) = \overline{F(\bar{z})}$, and $\Theta = E^\sharp/E$ (meromorphic on $\mathbb{C}$; $|\Theta| = 1$ on $\mathbb{R}$ off the zeros of E). Let $\Gamma_\zeta = \{\gamma_\rho\}$ be the multiset of zeros of A, i.e. $\rho = \frac{1}{2} - i\gamma_\rho$ runs over the nontrivial zeros of ζ; γ is real iff ρ lies on the critical line. Following [S2, (1.5)–(1.6), Prop. 3.1] define, unconditionally,

$$\mathfrak{P}_t(z) = \sum_\rho \frac{e^{-i\gamma t} - 1}{\gamma\,(z-\gamma)}, \qquad \mathfrak{S}_t = \frac{i(1+\Theta)}{2}\,\mathfrak{P}_t, \qquad G(t,u) = \sum_\rho \frac{(e^{i\gamma t} - 1)(e^{-i\gamma u} - 1)}{\gamma^2},$$

the sums over zeros with multiplicity, absolutely convergent ($\sum |\gamma|^{-2} < \infty$; $|e^{-i\gamma t} - 1| \le \min(|\gamma t|, 2)$ for real γ, with the obvious modification in horizontal strips). By [S2, Prop. 3.1] (the Weil explicit formula) $\mathfrak{P}_t$ coincides with the explicit zero-free expression [S2, (1.6)]; we work with the zero expansion. Note $\mathfrak{S}_t = iA\mathfrak{P}_t/E$ and $A\mathfrak{P}_t$ is entire: the poles of $\mathfrak{P}_t$ sit at Γ_ζ and are killed by A.

2 Functional-equation bookkeeping and the diagonal law

Lemma 1. *$A^\sharp = A$, $B^\sharp = -B$; hence $E^\sharp = A - B$, $A = (E + E^\sharp)/2$. Moreover $A' = -iB$. At a real zero γ of A with $B(\gamma) \neq 0$ (equivalently, ρ simple): $E(\gamma) = B(\gamma) \neq 0$, Θ is analytic in a neighborhood of γ, and $\Theta(\gamma) = -1$.*

Proof. Since ξ is real on $\mathbb{R}$, $\overline{\xi(\bar{w})} = \xi(w)$. Then $A^\sharp(z) = \overline{\xi(\frac{1}{2} - i\bar{z})} = \xi(\frac{1}{2} + iz) = \xi\big(1 - (\frac{1}{2} - iz)\big) = A(z)$ by the functional equation, and differentiating $\xi(1-s) = \xi(s)$ gives $\xi'(1-s) = -\xi'(s)$, whence $B^\sharp = -B$. The chain rule gives $A'(z) = -i\,\xi'(\frac{1}{2} - iz) = -iB(z)$. At γ: $E(\gamma) = A(\gamma) + B(\gamma) = B(\gamma)$ and $\Theta(\gamma) = (A-B)/(A+B)\big|_\gamma = -1$. $\square$

Lemma 2 (Diagonal law)**.** *At every simple real zero γ of A,*

$$i\,\Theta'(\gamma) = 2.$$

More generally, for any entire F, real on $\mathbb{R}$, with $F(1-s) = F(s)$, the same holds for $E_F = (F + F')(\frac{1}{2} - iz)$ at every simple real zero of $F(\frac{1}{2} - iz)$ at which $F' \neq 0$. No arithmetic input enters.

Proof. $\Theta' = (E^{\sharp\prime} E - E^\sharp E')/E^2$. At γ, $E^\sharp(\gamma) = -E(\gamma)$, so $\Theta'(\gamma) = \big(E^{\sharp\prime} + E'\big)(\gamma)/E(\gamma) = 2A'(\gamma)/E(\gamma) = -2iB(\gamma)/B(\gamma) = -2i$. $\square$

Remark 3. *Numerically $i\Theta'(\gamma_n) = 2$ to 10^{-16} at $n = 1, 2, 5$ (campaign file `att247b`/check). The law is forced by the functional equation — the* $\det(-1)$*-reflection normalizes the boundary derivative at every critical point; this is precisely the exactness that makes the Parseval below carry* unit *Clark weights.*

3 The Clark family

For a real zero γ (simple) define

$$u_\gamma(z) = \frac{i\,(1 + \Theta(z))}{2\,(z - \gamma)}.$$

Hypotheses. *(I)*: Θ is *inner* in $\mathbb{C}_+$, i.e. E has no zeros in $\mathbb{C}_+$ and $|\Theta| \leq 1$ on $\mathbb{C}_+$. *(S)*: all nontrivial zeros of ζ are simple. (By [S2, §2.3] and its reference for the Hermite–Biehler property, RH implies (I); conversely the Theorem below plus Weil positivity gives RH from (I). (S) is for bookkeeping only; see Remark 12.)

We use the model space $K_\Theta = H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+)$ with the $L^2(\mathbb{R})$ pairing $\langle f, g\rangle = \int_{\mathbb{R}} f\bar{g}\,dx$ and Szegő kernel $c_w(z) = \frac{i}{2\pi}(z - \bar{w})^{-1}$, so that K_Θ has reproducing kernel

$$k_w(z) = \frac{i}{2\pi} \cdot \frac{1 - \overline{\Theta(w)}\,\Theta(z)}{z - \bar{w}}, \qquad \langle f, k_w\rangle = f(w) \quad (f \in K_\Theta,\ w \in \mathbb{C}_+).$$

Lemma 4 (Boundary Clark kernels)**.** *Assume (I), (S). For each simple real zero γ:*

(a) $u_\gamma \in H^2(\mathbb{C}_+)$, with boundary values analytic near γ;

(b) $u_\gamma \perp \Theta H^2$, hence $u_\gamma \in K_\Theta$;

(c) $u_\gamma = \pi\,k_\gamma$, where $k_\gamma := \lim_{\varepsilon\downarrow 0} k_{\gamma+i\varepsilon}$ in $L^2(\mathbb{R})$;

(d) for $f \in K_\Theta$ whose boundary values extend continuously to a neighborhood of γ: $\langle f, k_\gamma\rangle = f(\gamma)$;

(e) $\langle u_\gamma, u_{\gamma'}\rangle = \pi\,\delta_{\gamma\gamma'}$.

Proof. (a) $1+\Theta \in H^\infty(\mathbb{C}_+)$ by (I). Near γ, E is zero-free on a $\mathbb{C}$-neighborhood (Lemma 1 and (I)), so Θ is analytic there and $1+\Theta$ vanishes at γ; hence $|u_\gamma(z)| \le C\min(1,|z-\gamma|^{-1})$ on $\overline{\mathbb{C}_+}$, giving $\sup_{y>0}\int_{\mathbb{R}} |u_\gamma(x+iy)|^2 dx < \infty$.

(b) For $h \in H^2$, on $\mathbb{R}$ we have $\overline{\Theta} = \Theta^{-1}$ and $\Theta\,\overline{u_\gamma} = \Theta \cdot \frac{-i(1+\overline{\Theta})}{2(z-\gamma)} = \frac{-i(\Theta+1)}{2(z-\gamma)} = \overline{\overline{\Theta+1}} \cdots$; concretely $\langle \Theta h, u_\gamma\rangle = \int_{\mathbb{R}} h \cdot \frac{-i(1+\Theta)}{2(z-\gamma)}\,dz$. The integrand is a product of $h \in H^2$ and $\frac{-i(1+\Theta)}{2(z-\gamma)} \in H^2$ (same argument as (a)), hence lies in $H^1(\mathbb{C}_+)$, and $\int_{\mathbb{R}} F\,dz = 0$ for $F \in H^1(\mathbb{C}_+)$.

(c) $\overline{\Theta(\gamma)} = -1$ by Lemma 1 (real point, real-type Θ). As $\varepsilon \downarrow 0$, $k_{\gamma+i\varepsilon}(z) = \frac{i}{2\pi}\big(1 - \overline{\Theta(\gamma+i\varepsilon)}\Theta(z)\big)(z-\gamma+i\varepsilon)^{-1} \to \frac{i}{2\pi}(1+\Theta(z))(z-\gamma)^{-1} = \pi^{-1}u_\gamma(z)$ pointwise; the family is dominated by $C\min(1,|z-\gamma|^{-1})$ uniformly in ε (analyticity of Θ near γ gives $|1-\overline{\Theta(\gamma+i\varepsilon)}\Theta(z)| \le C(|z-\gamma|+\varepsilon)$ there, and the denominator is $\ge \max(|z-\gamma|,\varepsilon)$), so the limit holds in $L^2(\mathbb{R})$ by dominated convergence.

(d) $\langle f, k_\gamma\rangle = \lim_\varepsilon \langle f, k_{\gamma+i\varepsilon}\rangle = \lim_\varepsilon f(\gamma+i\varepsilon) = f(\gamma)$, the last step by continuity of boundary values near γ (for $f \in K_\Theta \subset H^2$, nontangential limits agree with the continuous extension).

(e) For $\gamma \ne \gamma'$: $\langle u_{\gamma'}, u_\gamma\rangle = \pi\langle u_{\gamma'}, k_\gamma\rangle = \pi\, u_{\gamma'}(\gamma) = \pi \cdot \frac{i(1+\Theta(\gamma))}{2(\gamma-\gamma')} = 0$ since $\Theta(\gamma) = -1$. For $\gamma = \gamma'$: $\|u_\gamma\|^2 = \pi\, u_\gamma(\gamma) = \pi \lim_{z\to\gamma} \frac{i(1+\Theta(z))}{2(z-\gamma)} = \pi \cdot \frac{i\,\Theta'(\gamma)}{2} = \pi$, by the diagonal law (Lemma 2). □

4 The identity

Lemma 5 (Clark expansion of $\mathfrak{S}$)**.** *Set $c_\gamma(t) = (e^{-i\gamma t}-1)/\gamma$. Unconditionally, $\mathfrak{S}_t(z) = \sum_\rho c_\gamma(t)\,u_\gamma(z)$ pointwise for $z \notin \Gamma_\zeta$, and $\sum_\rho |c_\gamma(t)|^2 < \infty$. Under (I),(S) the series converges in $L^2(\mathbb{R})$ and its sum equals $\mathfrak{S}_t$ a.e.*

Proof. Pointwise: multiply the zero expansion of $\mathfrak{P}_t$ termwise by $i(1+\Theta)/2$. ℓ^2: $|c_\gamma| \le \min(|t|, 2/|\gamma|)$ and $\sum|\gamma|^{-2} < \infty$. L^2: by orthogonality (Lemma 4(e)) the partial sums are Cauchy: $\|\sum_{N<n\le M} c\,u\|^2 = \pi\sum_{N<n\le M}|c|^2 \to 0$. A sequence converging pointwise off a discrete set and in L^2 has matching limits a.e. □

Theorem 6. *Assume (I) and (S). Then for all real x, y,*

$$\frac{1}{\pi}\int_{\mathbb{R}} \mathfrak{S}_x(z)\,\overline{\mathfrak{S}_y(z)}\,dz \;=\; \sum_\rho c_\gamma(x)\overline{c_\gamma(y)} \;=\; G(x,y).$$

Consequently the Weil quadratic form is a sum of squares ($\pi\langle\psi,\psi\rangle_W = \|\widehat{\mathcal{P}}_{D\psi}\|^2_{L^2}$ in the notation of [S2]), and RH holds.

Proof. Parseval for the orthogonal family $\{u_\gamma\}$ with $\|u_\gamma\|^2 = \pi$ (Lemmas 4, 5): $\langle \mathfrak{S}_x, \mathfrak{S}_y\rangle = \pi\sum c_\gamma(x)\overline{c_\gamma(y)}$. Since γ is real under (I) (no zeros of A off $\mathbb{R}$: a nonreal zero $\gamma_0 \in \mathbb{C}_+$ of $A = (E+E^\sharp)/2$ would force $|\Theta(\gamma_0)| = |E^\sharp/E|(\gamma_0) = 1$ with Θ inner nonconstant, impossible unless $E(\gamma_0) = 0$, excluded by (I); zeros in $\mathbb{C}_-$ pair with $\mathbb{C}_+$ by $A^\sharp = A$), we have $\overline{c_\gamma(y)} = (e^{i\gamma y}-1)/\gamma$, and the $\gamma \mapsto -\gamma$ symmetry of the zero multiset identifies $\sum c_\gamma(x)\overline{c_\gamma(y)}$ with $G(x,y)$. The final implication is [S2, Thm. 1.2] (positivity direction: immediate, since the left side is a norm). □

5 Status of the hypothesis, and the defect program

Proposition 7 (Cayley identification)**.** *In s-coordinates write* $w(s) = \xi'/\xi(s)$*, so that* $\Theta = (1-w)/(1+w)$*. Then* $|\Theta| \le 1$ *at a point of* $\mathbb{C}_+$ *iff* $\operatorname{Re} w \ge 0$ *at the corresponding s with* $\operatorname{Re} s > \frac{1}{2}$*, and the* $\mathbb{C}_+$*-poles of* Θ *are the zeros of* $1+w$ *(the points* $\xi'/\xi = -1$*; at zeros of* ξ *one has* $w = \infty$*,* $\Theta = -1$*, no pole). Hence*

$$\text{(I)} \iff \operatorname{Re} \frac{\xi'}{\xi}(s) \;\ge\; 0 \quad \textit{for all } \operatorname{Re} s > \tfrac{1}{2},$$

i.e. the hypothesis is exactly the positivity of the field $S(s) = 2\operatorname{Re}[\xi'/\xi]/(2\sigma - 1)$ *of the scalar seat criterion (RH_LEDGER 218), the Hinkkanen–Lagarias positivity.*

Lemma 8 (Exact pair law)**.** *Group the nontrivial zeros into functional-equation pairs* $\{\beta + i\gamma_c, 1-\beta+i\gamma_c\}$ *and write* $a = \beta - \frac{1}{2}$*,* $x = \sigma - \frac{1}{2} > 0$*,* $\Delta = t - \gamma_c$*. From* $\operatorname{Re} \xi'/\xi(s) = \sum_\rho (\sigma - \beta_\rho)/|s-\rho|^2$ *(Hadamard), each pair contributes exactly*

$$\frac{2x\left[x^2 - a^2 + \Delta^2\right]}{D_1 D_2}, \qquad D_{1,2} = (x \mp a)^2 + \Delta^2.$$

This is negative only when $|\Delta| < \sqrt{a^2 - x^2} \le |a| \le \frac{1}{2}$*: a pair can push the field negative only inside a disk of radius at most* $\frac{1}{2}$ *about its own height, and only at abscissae* $x < |a|$*.*

Proof. $(x-a)D_2 + (x+a)D_1 = (x^2 - a^2)\cdot 2x + 2x\Delta^2$; the constraint $|a| \le \frac{1}{2}$ is the critical strip. □

Theorem 9 (The hypothesis below the verified edge)**.** *Let* T_0 *be a height below which all nontrivial zeros are on the critical line (verified computation;* $T_0 = 3 \cdot 10^{12}$ *suffices). Then for every* $\sigma > \frac{1}{2}$ *and* $|t| \le T_0 - \frac{1}{2}$*,*

$$\operatorname{Re} \frac{\xi'}{\xi}(\sigma + it) \;>\; 0,$$

term by term in the pair decomposition. Consequently $E = \xi + \xi'$ *is zero-free and* $|\Theta| < 1$ *there: hypothesis (I) holds on the entire half-plane strip below the verified edge, with buffer* $\frac{1}{2}$ *and no low-height exclusion.*

Proof. A pair with $|\Delta| < \frac{1}{2}$ has $|\gamma_c| \le |t| + \frac{1}{2} \le T_0$, hence is on-line ($a = 0$) and contributes $2x/D > 0$. A pair with $|\Delta| \ge \frac{1}{2}$ has $\Delta^2 \ge \frac{1}{4} \ge a^2$, so $x^2 - a^2 + \Delta^2 \ge x^2 > 0$. Every term is positive. □

Remark 10 (Co-locality law above the edge)**.** *By Lemma 8, any failure of (I) at height t — in particular any* $\mathbb{C}_+$*-zero of E — requires an off-critical zero within distance* $\frac{1}{2}$ *of t, with displacement* $|a|$ *exceeding the abscissa x of the failure; and it must overcome the positive on-line mass* $\sum 2x/(x^2 + \Delta_j^2) \sim \log t$ *at that height. The seat's failure and its cause are co-local. Combined with the local converse of RH_LEDGER 222 (negativity within distance 1 of an off-line zero), the remaining content of (I) is a per-height local dichotomy above* T_0*. Numerics: the pair-sum representation reproduces* $\operatorname{Re} \xi'/\xi$ *to* 10^{-8} *at spot points, and the measured floor at* $x = 10^{-2}$*,* $t \in [40, 60]$ *is* $+5.7 \cdot 10^{-3}$ *(`att250`); the* $\mathbb{C}_+$ *census and control are in `att246`.*

Remark 11 (Defect coordinates)**.** *Without (I), Lemma 4 fails exactly through the $\mathbb{C}_+$-poles of Θ: the reproducing computation acquires residue corrections at the $\mathbb{C}_+$-zeros of E, so $\pi G(x,y) - \langle \mathfrak{S}_x, \mathfrak{S}_y \rangle$ is a quadratic expression supported on those zeros. Deriving this defect formula explicitly (and its positivity structure) is the quantitative continuation.*

Remark 12 (Multiplicities)**.** *At an m-fold real zero, $\Theta(\gamma) = -1$ still, while $\Theta'(\gamma) = 2A'(\gamma)/E(\gamma)$ requires the obvious modification ($A', \ldots, A^{(m-1)}$ vanish); the Clark family acquires the standard derivative kernels and the expansion coefficients the corresponding jets. The bookkeeping is routine and not needed under (S).*

6 Boundedness from zero-freeness

Theorem 13 ((Z) $\Rightarrow$ RH)**.** *Suppose $E = \xi + \xi'$ (in the z-chart, $E(z) = A(z) + B(z)$) has no zeros in the open band $\{\operatorname{Re} z > T_0 - \frac{1}{2},\ 0 < \operatorname{Im} z < \frac{1}{2}\}$ — equivalently, by the previously proven regions, none in $\mathbb{C}_+$. Then $|\Theta| \le 1$ on $\mathbb{C}_+$ and RH holds. In particular the boundedness hypothesis of the band theorem is not independent: it follows from zero-freeness.*

Proof. Under (Z): $\Theta = E(-z)/E(z)$ is analytic in $\mathbb{C}_+$ and continuous up to $\mathbb{R}$ with $|\Theta| = 1$ there (real E-zeros are common to $E, E^\sharp$ with equal multiplicity, hence removable); $h = \log|\Theta|$ is subharmonic on the band.

Step 1 (anchor at $\sigma = 2$). Let $z_c = t + \frac{3}{2}i$, i.e. $s = 2 - it$. Then $E(z_c) = \xi(2-it)\,(1 + w(2-it))$ with $|1 + w| > 1$ *free* from the termwise lemma ($\operatorname{Re} w > 0$ for $\sigma \ge 1$), and $|\xi(2-it)| \ge e^{-\pi t/4} t^{-C}/\zeta(2)$ by Stirling. Hence $\log|E(z_c)| \ge -\pi t/4 - C\log t$ unconditionally. (Measured: $|E(z_c)|e^{\pi t/4}$ grows polynomially, $2.6 \cdot 10^3 \to 8.0 \cdot 10^4$ over $t = 30 \to 100$; $|1+w| \in [2.2, 2.5]$; `att252`.)

Step 2 (local count). Jensen on $D(z_c, 5.8)$ against the anchor and the Stirling envelope $\log|E| \le -\pi t/4 + C\log t$ (valid on horizontal strips of bounded width, for ξ and ξ' alike) gives $n := \#\{E\text{-zeros in } D(z_c, 2.9)\} \le C\log t$. No circularity: only the anchor and the envelope enter.

Step 3 (minimum modulus on the band). Write $E = \prod_{j \le n}(\,\cdot\, - e_j)\,g$ with g zero-free on $D(z_c, 2.9)$. Borel–Carathéodory/Harnack applied to the positive harmonic function $\max_{D(2.9)} \log|g| - \log|g|$, anchored at z_c, gives $\log|g| \ge -\pi t/4 - C(\log t)^2$ on $D(z_c, 2.4)$. At a band point $z = t' + iy$ ($|t' - t| \le 1$, $0 < y < \frac{1}{2}$) the zero factors cost $\sum_j \log|z - e_j| \ge n \log y$, since under (Z) every e_j has $\operatorname{Im} e_j \le 0$ while z sits at height y. With the upper envelope for $|E(-z)|$:

$$h(z) = \log|\Theta(z)| \;\le\; C(\log t)^2 + C\log t\,\log(1/y).$$

Step 4 (corner cap). For $y \le 1/t$ the local maximum principle on the half-disk $D((t', 0), 2/t)$ — boundary values 0 on the diameter, $\le C(\log t)^2$ on the arc by Step 3 — caps $\sup_{0<y<1/2} h(t' + iy) \le C(\log t)^2$ *uniformly*: the $\log(1/y)$ blowup is an artifact of the estimate, not of h.

Step 5 (strip Phragmén–Lindelöf). h is subharmonic on the half-strip of width $\frac{1}{2}$ with $\limsup h \le 0$ at every finite boundary point (bottom: continuity, $h = 0$ on $\mathbb{R}$; top $\sigma = 1$ and left edge: the proven regions) and growth $C(\log t)^2$, far below the allowance $\exp(\exp((2\pi - \varepsilon)t))$; comparison with $\varepsilon \operatorname{Re} e^{c(z - T_0)}$-type majorants gives $h \le 0$ on the band, hence $|\Theta| \le 1$ on all of $\mathbb{C}_+$.

Step 6 (endgame). At a hypothetical off-line A-zero, $\Theta = -1$ is an interior modulus maximum, so $\Theta \equiv -1$, i.e. $\xi' \equiv 0$, absurd. Hence no off-line zeros: RH. All remaining constants are Stirling-envelope bookkeeping. $\square$

Remark 14 (Phase positivity, no longer load-bearing)**.** *The earlier corner guard via the phase derivative is subsumed by Step 4; the facts remain true and instrument-grade:* $\varphi'(t) = -\operatorname{Im}(E'/E)(t)$ *measured positive at all sampled heights (*0.30–1.12*), and at an on-line zero* $\varphi'(\gamma) = |\Theta'(\gamma)|/2 = 1$ exactly *— the diagonal law as unit boundary phase velocity (measured* 0.99997 *at* γ_1*;* `att251`*).*

Remark 15 (Both directions constructive)**.** *Conversely, if all zeros above* T_0 *are on the critical line, the pair law makes* $\operatorname{Re} \xi'/\xi > 0$ *termwise at every height, so (Z) holds. Thus (Z)* $\iff$ *RH with both directions constructive, and a band E-zero is enslaved to an off-critical* ξ*-zero within distance* $\frac{1}{2}$*: on any height where the local zeros are on-line,* $\operatorname{Re} w \geq 0$ *and the value* -1 *is unreachable.*

Remark 16 (Final coordinates of RH)**.** *Combining:* RH $\iff$ $\xi'/\xi(s) \neq -1$ for all $\sigma > \frac{1}{2}$, $t > T_0$ *— one equation's avoidance of one band. All other regions are settled unconditionally; boundedness is derived, not assumed; and the two remaining technical lemmas are classical-methods bookkeeping, stated above at their exact strength.*

Proposition 17 (Phase–Laguerre bridge)**.** *On* $\mathbb{R}$*, with* $\Xi(t) = \xi(\frac{1}{2} + it) = A(t)$ *and* $|E|^2 = \Xi^2 + \Xi'^2$*,*

$$\varphi'(t) \;=\; \frac{\Xi'(t)^2 - \Xi(t)\Xi''(t)}{\Xi(t)^2 + \Xi'(t)^2}.$$

Thus the boundary phase monotonicity of E is exactly the Laguerre/Turán inequality for Ξ *on the critical line — the unconditional target of the theta–Wronskian campaign (RH_LEDGER 238–262). At an on-line zero the numerator is* $\Xi'(\gamma)^2 = |E(\gamma)|^2$*, recovering the diagonal law* $\varphi'(\gamma) = 1$*. Verified to ten digits at* $t = 20, 45.75, 77$ *(*`att253`*).*

Remark 18 (The residual, in counting form)**.** *Phase monotonicity is necessary but not sufficient for (Z): a* $\mathbb{C}_+$*-zero of E at height t removes* 2π *of phase across its Poisson width, which ambient density* $\sim \log t$ *can absorb without violating* $\varphi' \geq 0$*. What remains beyond the Laguerre inequality is* saturation*: the phase increment over each window must account* π *per E-zero as lying below* $\mathbb{R}$*. Equivalently* $(Z) \Leftarrow$ *[Laguerre(*Ξ*)* ≥ 0*] + [per-window phase-count saturation], the latter an integer-valued, census-friendly statement — the effective per-height form of the remaining core.*

Remark 19 (Numerical control)**.** *All exact constants above were verified numerically in the campaign files:* $i\Theta'(\gamma) = 2$ *to* 10^{-16}*;* $\|u_{\gamma_1}\|^2, \|u_{\gamma_2}\|^2 \to \pi$ *and* $\langle u_{\gamma_1}, u_{\gamma_2}\rangle \to 0$ *under truncation (*`att249`*); the identity itself at three* (x, y) *pairs to the truncation tail (*`att247b`*,* `att248b`*), with* $\mathfrak{P}$ *validated against its zero expansion at* 10^{-12}*.*